ON ORTHOMORPHOSIS (continued)

[920, pp. 179–190]

and the two equations give $x^2 + y^2 - 1 = 0$; the two circumferences thus correspond to each other. It is to be noticed that, if the numerator function contain a factor $(x + iy)^m$, say if $\phi(x + iy) = (x + iy)^m \psi(x + iy)$, then the denominator will be

$$(x + iy)^{-m}\, \overline{\psi}\left(\frac{1}{x + iy}\right),$$

and the formula thus becomes

$$x_1 + iy_1 \;=\; \frac{(x + iy)^m\, \psi(x + iy)}{\dfrac{1}{(x + iy)^m}\, \overline{\psi}\left(\dfrac{1}{x + iy}\right)},$$

which is thus not really more general than the original form. I recur further on to this transformation between the two circumferences.

10. The curve S may be taken to be a closed curve enclosing a singly connected area F (say such a curve is a Contour), and the curve S' a like curve lying wholly inside S, and such that the normal distance between the two curves is everywhere infinitesimal. It is not in general the case, but it may happen that the successive curves S'', S''', &c. will be all of them like curves (each enclosed in the preceding and enclosing the succeeding one), of continually diminishing area, and continually approximating to a point: the curves T will then all of them pass through this point and may be called Radials. We may say that the area F is then squarewise contractible, or contracted, into a point.

11. In particular, a circle is squarewise contractible into a point, viz. if the point be the centre, then the curves S are concentric circles, and the curves T are radii; as is known and will appear, the circle is also contractible into any interior eccentric point whatever. It may be remarked that (the squares being infinitesimal) the numbers of the contours and radials are each of them infinite, but further the number of contours is infinitely great in comparison with that of the radials. Thus in the case just referred to, if to construct a figure we imagine the circumference of the circle divided into any large number n of equal parts, the radials will be the n radii drawn to the points of division: taking the radius to be $= 1$, and writing for shortness $\alpha = \dfrac{2\pi}{n}$, the sides of the successive squares along any radius will be α, $\alpha(1-\alpha)$, $\alpha(1-\alpha)^2, \ldots$, and we require an infinite number of these to make up the entire radius; $1 = \alpha\{1 + (1 - \alpha) + (1 - \alpha)^2 + \ldots\}$; viz. the number of radials being any large number n whatever, the number of contours will be actually infinite.

12. We may compare the circle as thus contracted into its centre, or rather the quadrant of a circle, with an infinite strip of finite breadth divided into squares by two sets of parallel orthotomic lines: here if x_1, y_1 refer to the positive quadrant of the circle $x_1^2 + y_1^2 = 1$, and x, y to the infinite strip $x = 0$ to $-\infty$, $y = 0$ to $\frac{1}{2}\pi$, we have $x + iy = \log(x_1 + iy_1)$, that is,

$$x = \log\sqrt{(x_1^2 + y_1^2)}, \qquad y = \tan^{-1}\frac{y_1}{x_1};$$

the successive concentric quadrants correspond to equal lines parallel to the axis of y, and the successive radii to infinite lines parallel to the axis of x.

13. Considering the contour S as given, then for a consecutive interior contour S' assumed at pleasure, the area F does not contract into a point: but we have the theorem that the consecutive contour S' can be found, and that in one way only, such that the area F shall contract into a given interior point. To do this is, in effect, to make the given area to correspond say to the circle $x_1^2 + y_1^2 - 1 = 0$, the contour S and the successive interior contours to the circumference of the circle and to the circumferences of the successive interior concentric circles, and finally the given interior point of F to the centre $x_1 = 0$, $y_1 = 0$ of the circle: and thus the theorem is identical with Riemann's theorem "It is possible and that in one way only to make a given singly connected area F correspond to a circle, in such wise that a given interior point of F shall correspond to the centre of the circle, and that a given boundary point of F shall correspond to a given point on the circumference of the circle." The last clause as to the given boundary point of F is necessary in order to make the correspondence a completely definite one, for without the clause it would obviously be allowable to give to the circle an arbitrary rotation about its centre.

14. The proof depends on the following lemma[1].

For the given area F, it is possible to find, and that in one way only, a real function ξ of the coordinates (x, y) satisfying the following conditions:

(1) ξ is throughout the area finite and continuous, except only that in the neighbourhood of a given point thereof, taken to be the point $x = 0$, $y = 0$, it is $= \log\sqrt{(x^2 + y^2)}$.

(2) At the boundary of the area ξ is $= 0$.

(3) Throughout the area ξ satisfies the partial differential equation

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0.$$

In fact, if ξ be thus determined, it follows from (3) that we have

$$-\frac{d\xi}{dy}\,dx + \frac{d\xi}{dx}\,dy = d\eta,$$

an exact differential, whence putting this $= d\eta$, or determining η by the quadrature

$$\eta = \int\left(-\frac{d\xi}{dy}\,dx + \frac{d\xi}{dx}\,dy\right),$$

we have

$$\frac{d\eta}{dx} = -\frac{d\xi}{dy}, \qquad \frac{d\eta}{dy} = \frac{d\xi}{dx},$$

and thence $\xi + i\eta$ is a function of $x + iy$.

Writing then $x_1 + iy_1 = e^{\xi + i\eta} = \psi(x + iy)$ suppose, we have an orthomorphosis of the area into the circle $x_1^2 + y_1^2 - 1 = 0$, viz. for any point (x, y) of the boundary $\xi = 0$, and consequently $x_1 + iy_1 = e^{i\eta}$, thence $x_1 - iy_1 = e^{-i\eta}$, and therefore $x_1^2 + y_1^2 - 1 = 0$, or the point corresponds to a point on the circumference of the circle. Also, for a point x, y in the neighbourhood of $x = 0$, $y = 0$, we have

$$x_1 + iy_1 = e^{\log\sqrt{(x^2+y^2)} + i\eta} = \sqrt{(x^2 + y^2)}\,e^{i\eta},$$

that is, for $x = 0$, $y = 0$ we have $x_1 = 0$, $y_1 = 0$, or the given point of F corresponds to the centre of the circle.

Some further considerations as to the continuity of η would be requisite in order to show that to the series of circles $x_1^2 + y_1^2 = c^2$, as c continuously increases from 0 to 1, there correspond closed curves surrounding the assumed origin and each other, and passing continuously to the boundary of F, and thus to complete the proof that we thus obtain an orthomorphosis of F into the circle $x_1^2 + y_1^2 = 1$; but I abstain from a further discussion of the question.

15. Reverting to the lemma, this in the first place asserts the existence of a function ξ satisfying the prescribed conditions, and next that the function is completely determinate, or say that there is but one such function. As to the latter point, suppose that there is a second function ξ_1 satisfying the same conditions, then throughout the area $\xi - \xi_1$ is finite and continuous; by general principles in the theory of functions, this implies that the function has throughout

[1]This proof is taken from the paper, Christoffel, "Sul problema delle temperature stazionarie et la rappresentazione di una data superficie," *Annali di Matem.*, t. i. (1867), pp. 89—103.

the area a constant value, and since this value at the boundary is $= 0$, it must be always $= 0$, that is, $\xi - \xi_1 = 0$, or $\xi_1 = \xi$. As to the former point, it is to be remarked that we obtain

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0$$

as the general condition for a minimum value of the double integral

$$\iint \left\{ \left(\frac{d\xi}{dx}\right)^2 + \left(\frac{d\xi}{dy}\right)^2 \right\} dx\, dy;$$

if then we assume that there exists a function ξ, $= 0$ at the boundary of the area, and finite and continuous throughout the area, except only that in the neighbourhood of a given point thereof, say $x = 0$, $y = 0$, it becomes $= \log \sqrt{(x^2 + y^2)}$, and that for all the values which satisfy these conditions it is such that the above-mentioned double integral taken over the given area shall be a minimum, it follows that there exists a function ξ satisfying the conditions of the lemma.

16. As a simple illustration, suppose that the area F is that of the circle $x^2 + y^2 = 1$, and that the excepted point is the point $x = 0$, $y = 0$, the centre of the circle. We have here a function ξ satisfying the conditions of the lemma, viz. this is $\xi = \log \sqrt{(x^2 + y^2)}$; the resulting value of η is $\eta = \tan^{-1} \dfrac{y}{x}$, and we have then $\xi + i\eta$ a function of $x + iy$, viz. this is $= \log(x + iy)$. Hence $\xi + i\eta = x + iy$, and the resulting orthomorphosis of the two circles is the identical one

$$x_1 + iy_1 = x + iy,$$

viz. we have

$$x_1 = x, \qquad y_1 = y.$$

17. The contraction of a circle to a given eccentric point has, in fact, been exhibited in the foregoing example in No. 5, but I resume the question from a different point of view. Writing for shortness $z_1 = x_1 + iy_1$, $z = x + iy$; using also a bar for the conjugate function, $\bar{z} = x - iy$, and so in other cases $\alpha = a + a'i$, $\bar{\alpha} = a - a'i$, $\beta = b + b'i$, &c., the general formula of (7) may be written in the form

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha} z} \cdot \frac{z - \beta}{1 - \bar{\beta} z} \cdots,$$

viz. we thus have a transformation between the circumferences $x^2 + y^2 - 1 = 0$, $x_1^2 + y_1^2 - 1 = 0$. In fact, repeating the verification, the change of sign of i gives

$$\bar{z}_1 = \frac{\bar{z} - \bar{\alpha}}{1 - \alpha \bar{z}} \cdot \frac{\bar{z} - \bar{\beta}}{1 - \beta \bar{z}} \cdots$$

Here the product of the α-factors is

$$\frac{z\bar{z} - (\alpha\bar{z} + \bar{\alpha} z) + \alpha\bar{\alpha}}{1 - (\alpha z + \bar{\alpha}\bar{z}) + \alpha\bar{\alpha} z\bar{z}},$$

which, if $z\bar{z} = 1$, becomes $= 1$; the same property exists as to the β-factors, &c., and hence putting $z\bar{z} = 1$, we have $z_1\bar{z}_1 = 1$, that is, the two circumferences correspond to each other. The correspondence would have subsisted if a factor $e^{i\kappa}$ had been introduced into the expression of z_1, but the effect is merely to rotate the circle $x_1^2 + y_1^2 - 1 = 0$ about its centre, and there is no real gain of generality.

18. The most simple case is

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha} z}.$$

I assume here that $\alpha = a + a'i$ is an interior point of the circle $x^2 + y^2 - 1 = 0$ (viz. $a^2 + a'^2 < 1$); hence $z = \alpha$ gives $z_1 = 0$, viz. to the point $x = a$, $y = a'$, within the circle $x^2 + y^2 - 1 = 0$, there corresponds the centre of the circle $x_1^2 + y_1^2 - 1 = 0$; and we have in this equation the theory of the squarewise contraction of the circle to the eccentric point $x = a$, $y = a'$. There is no loss of generality in assuming $a' = 0$; doing this the formula becomes

$$z_1 = \frac{z - a}{1 - az},$$

where $a^2 < 1$, or if to fix the ideas a be taken to be positive, then $a < 1$.

19. To compare with a former investigation, writing $\xi + i\eta = \log(x_1 + iy_1)$, we have

$$\xi + i\eta = \log\sqrt{((x-a)^2 + y^2)} + i\tan^{-1}\frac{y - ay}{x - a + a(x^2 + y^2)},$$

and consequently

$$\xi = \tfrac{1}{2}\log\frac{(x-a)^2 + y^2}{(1-ax)^2 + a^2y^2},$$

which is $= 0$ at the boundary $x^2 + y^2 - 1 = 0$, and is finite and continuous throughout the area except in the neighbourhood of the point $x = a$, $y = 0$, for which it is $= \log\sqrt{((x-a)^2 + y^2)}$; moreover, ξ satisfies the partial differential $\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0$, and starting from the given value of ξ we should obtain the foregoing value of η, and thence $x_1 + iy_1 = e^{\xi + i\eta}$, that is,

$$x_1 + iy_1 = \frac{z-a}{1-az}, \quad \text{or} \quad z_1 = \frac{z-a}{1-az},$$

as the equation for the orthomorphosis of the circle $x^2 + y^2 - 1 = 0$ into the circle $x_1^2 + y_1^2 - 1 = 0$, the centre $x_1 = 0$, $y_1 = 0$ corresponding to the eccentric point $x = a$, $y = 0$.

20. In further development of the solution, writing

$$z_1\bar{z}_1 = \frac{z\bar{z} - a(z + \bar{z}) + a^2}{1 - a(z + \bar{z}) + a^2 z\bar{z}},$$

that is,

$$x_1^2 + y_1^2 = \frac{x^2 + y^2 - 2ax + a^2}{a^2(x^2 + y^2) - 2ax + 1},$$

and then assuming $x_1^2 + y_1^2 = \lambda^2$, we have as the contour corresponding thereto

$$x^2 + y^2 + a^2 - 2ax = \lambda^2\{a^2(x^2 + y^2) - 2ax + 1\},$$

that is,

$$(x^2 + y^2)(1 - a^2\lambda^2) - 2a(1 - \lambda^2)x + a^2 - \lambda^2 = 0,$$

or say

$$x^2 + y^2 - \frac{2a(1-\lambda^2)}{1 - a^2\lambda^2}x + \frac{a^2 - \lambda^2}{1 - a^2\lambda^2} = 0,$$

or, what is the same thing,

$$\left(x - \frac{a(1-\lambda^2)}{1 - a^2\lambda^2}\right)^2 + y^2 = \frac{\lambda^2(1-a^2)^2}{(1-a^2\lambda^2)^2}.$$

The equation may also be written

$$\left\{x - \frac{a(1-\lambda^2)}{1 - a^2\lambda^2}\right\}^2 + y^2 = \frac{\lambda^2(1-a^2)^2}{(1-a^2\lambda^2)^2},$$

showing that the contour circles all pass through two imaginary points

$$x = \frac{1}{a}\left(1 - \frac{1}{a}\right), \qquad y = \pm\frac{i}{a}\left(\frac{1}{a} - a\right).$$

21. Writing next

$$z_1 = \frac{(z-a)(1-a\bar{z})}{(1-az)(1-a\bar{z})} = \frac{(z-a)(1-a\bar{z})}{|1-az|^2},$$

that is,

$$x_1 = \frac{-a + x(1+a^2) - a(x^2+y^2)}{(1-ax)^2 + a^2y^2}, \qquad y_1 = \frac{y(1-a^2)}{(1-ax)^2+a^2y^2},$$

then to the radius $x_1 - \theta y_1 = 0$ corresponds the radial

$$-a + x(1+a^2) - a(x^2+y^2) - \theta y(1-a^2) = 0,$$

that is,

$$(x-a)\left(x - \frac{1}{a}\right) + y^2 + \theta y\left(\frac{1}{a} - a\right) = 0,$$

which is a circle passing through the two real points $y = 0$, $x = a$, or $\frac{1}{a}$, being the antipoints of the before-mentioned pair of points.

22. Hence for the contraction of the circle $x^2 + y^2 - 1 = 0$ to the interior point $x = a$, $y = 0$, calling this point A, and taking A' the image hereof (coordinates $x = \frac{1}{a}$, $y = 0$), we see that the contour circles are the circles all passing through the antipoints of A, A', and having thus for their common chord the line bisecting AA' at right angles, and that the radial circles are the circles all passing through the two points A, A'.

In what precedes, the distance AA' is $= \frac{1}{a} - a$, or say the half distance is $= \frac{1}{2}\left(\frac{1}{a} - a\right)$; but altering the scale so as to make this half distance $= 1$, and moreover taking the origin at the middle point of AA', the equations of the contour circles and the radial circles assume the forms

$$x^2 + y^2 - 2\alpha x + 1 = 0,$$

and

$$x^2 + y^2 - 2\beta y - 1 = 0,$$

respectively.

23. It is better, interchanging x, y and altering the constants, to write these in the forms

$$x^2 + y^2 - \left(b + \frac{1}{b}\right) y + 1 = 0,$$

$$x^2 + y^2 - \left(a - \frac{1}{a}\right) x - 1 = 0,$$

viz. as in effect appearing in No. 5, these are the equations derived from $x + iy = \tan(u + iv)$ where $\tan u = a$, $\tan iv = ib$.

24. Passing now to the form

$$z_1 = \frac{z-\alpha}{1-\bar{\alpha}z} \cdot \frac{z-\beta}{1-\bar{\beta}z},$$

to obtain the most simple instance, I write $\alpha = 0$, $\beta = \sqrt{2}$, so that the form is

$$z_1 = z \cdot \frac{z - \sqrt{2}}{1 - \sqrt{2}\,z};$$

we thus obtain
that is,

$$x_1 + iy_1 = \frac{(x^2+y^2)(x^2+y^2-2\sqrt{2}\,x+2)}{2(x^2+y^2)-2x\sqrt{2}+1},$$

and thence

$$x_1^2 + y_1^2 = \frac{(x^2+y^2)\{x^2+y^2-2\sqrt{2}\,x+2\}}{2(x^2+y^2)-2x\sqrt{2}+1},$$

so that the circumference $x_1^2+y_1^2-1=0$ corresponds to the two circumferences $x^2+y^2-1=0$, $x^2+y^2-2\sqrt{2}\,x+1=0$. The second of these is $(x-\sqrt{2})^2+y^2-1=0$; hence, putting $x+\dfrac{1}{\sqrt{2}}$ instead of x, the two circles are

$$\left(x+\frac{1}{\sqrt{2}}\right)^2+y^2-1=0, \qquad \left(x-\frac{1}{\sqrt{2}}\right)^2+y^2-1=0,$$

(the former of these being the circle originally represented by $x^2+y^2-1=0$), and the last equation becomes

$$x_1^2+y_1^2-1 = \frac{\left\{\left(x+\frac{1}{\sqrt{2}}\right)^2+y^2-1\right\}\left\{\left(x-\frac{1}{\sqrt{2}}\right)^2+y^2-1\right\}}{2(x^2+y^2)}.$$

25. Writing here $x_1^2+y_1^2-c^2=0$, we have

$$\left\{\left(x+\frac{1}{\sqrt{2}}\right)^2+y^2-1\right\}\left\{\left(x-\frac{1}{\sqrt{2}}\right)^2+y^2-1\right\} = 2c^2(x^2+y^2)(1-c^2)=0,$$

that is,

$$(x^2+y^2)^2-(2c^2+1)x^2-(2c^2-1)y^2+\tfrac{1}{4}=0,$$

a bicircular quartic which (or rather a branch whereof) is the contour corresponding to the circle $x_1^2+y_1^2-c^2=0$. For $c=1$, the curve breaks up into the two circles

$$\left(x+\frac{1}{\sqrt{2}}\right)^2+y^2-1=0, \qquad \left(x-\frac{1}{\sqrt{2}}\right)^2+y^2-1=0,$$

and for $c=0$, it breaks up into

$$\left(x+\frac{1}{\sqrt{2}}\right)^2+y^2=0, \qquad \left(x-\frac{1}{\sqrt{2}}\right)^2+y^2=0,$$

that is, into the two points $y=0$, $x=\pm\dfrac{1}{\sqrt{2}}$. We have to consider the curves for values of c between these two values $c=1$ and $c=0$.

26. For any such value of c, the curve consists of two symmetrical ovals, situate within the two circles respectively: we are concerned only with the oval lying within the circle $\left(x+\dfrac{1}{\sqrt{2}}\right)^2+y^2-1=0$; this is shown in the figure for a value of c slightly inferior to 1, viz. it is an indented oval close approximating to the unshaded portion of the circle, say this is the contour S' consecutive to the circle which is the contour S.

As c diminishes and becomes ultimately $=0$, this oval continually diminishes,

after a time losing the indentation, and approximating more and more to the form of a circle having for its centre the point $x=-\dfrac{1}{\sqrt{2}}$, $y=0$, which is the centre of the circle, and ultimately reducing itself to this point.

It will be noticed that the consecutive contour S' as shown in the figure is not at an infinitesimal distance from every part of the circle which is the contour S; for a part of the circle, the consecutive contour or contour at an infinitesimal distance is a part of the symmetrically situated oval, lying outside of the circle. And thus in the present case we do not obtain a contraction of the circle into its centre: what we do obtain is a contraction of the unshaded portion of the circle into the centre; this is as it should be, for the only contraction of the circle into its centre is by the series of concentric circles.

27. For the radials, writing as before $x + \dfrac{1}{\sqrt{2}}$ in place of x, that is, $z + \dfrac{1}{\sqrt{2}}$ in place of z, we have

$$z_1 = z + \frac{1}{\sqrt{2}} \cdot \frac{\left(z + \frac{1}{\sqrt{2}}\right)\left(z - \frac{1}{\sqrt{2}}\right)}{-z\sqrt{2}} = -\frac{(z + \frac{1}{\sqrt{2}})(z^2 - \frac{1}{2})}{z\sqrt{2}}.$$

Hence

$$x_1 = -x - \frac{\frac{1}{2}x}{x^2+y^2} \cdot \frac{x(x^2+y^2-\frac{1}{2})}{x^2+y^2}, \qquad y_1 = -y - \frac{\frac{1}{2}y}{x^2+y^2} \cdot \frac{y(x^2+y^2+\frac{1}{2})}{x^2+y^2},$$

and thus to the radius $x_1 - \theta y_1 = 0$ there corresponds the curve

$$x(x^2+y^2-\tfrac{1}{2}) - \theta y(x^2+y^2+\tfrac{1}{2}) = 0,$$

or say

$$(x - \theta y)(x^2+y^2) - \tfrac{1}{2}(x + \theta y) = 0,$$

a circular cubic. In particular, for the line $x_1 = 0$, we have $\theta = 0$, and the curve is $x(x^2+y^2-\frac{1}{2}) = 0$, or say $x^2+y^2-\frac{1}{2} = 0$, the half circumference of which is shown in the figure: for the line $y_1 = 0$, we have $\theta = \infty$, and the curve is $y(x^2+y^2+\frac{1}{2}) = 0$, that is, $y = 0$.

The curve passes through the origin $x = 0$, $y = 0$, and the equation of the tangent there is $x + \theta y = 0$. Moreover it passes through the centre of the circle, $x = -\dfrac{1}{\sqrt{2}}$, $y = 0$, and the equation of the tangent there is easily found to be

$$x + \frac{1}{\sqrt{2}} - \theta y = 0.$$

We may find where the curve cuts the circle $\left(x + \dfrac{1}{\sqrt{2}}\right)^2 + y^2 - 1 = 0$, viz. this is $x^2+y^2 = -x\sqrt{2} + \frac{1}{2}$, and substituting in the equation of the curve, we find

$$x^2 - \theta xy + \frac{\theta}{\sqrt{2}}y = 0,$$

or say

$$\left(x - \frac{1}{\sqrt{2}}\theta\right)\left(x - \theta y + \frac{1}{\sqrt{2}}\right) + \tfrac{1}{2} = 0,$$

a hyperbola which by its intersections with the circle determines the points in question. There are two real intersections, but of these only one is in the arc bounding the unshaded portion of the circle: the other intersection serves to determine by symmetry the intersection of the circular cubic with the other boundary of the unshaded portion: see the figure which exhibits the path of the circular cubic through the unshaded portion, and into and beyond the shaded portion or lens. I remark that the two systems of curves are considered by Meyer and shown in his Plate XIII.

28. The formula

$$z_1 = \frac{z - \alpha}{1 - \bar{\alpha}z}$$

has been considered for the case of an interior point α; the case of an exterior point α might be considered in like manner, but it is obvious that we pass from one to the other, by a transformation $\dfrac{1}{z_1}$ for z_1; and it thus easily appears that, if α be an exterior point, we obtain a transformation between the area of the circle $x_1^2+y_1^2-1=0$ and the infinite area exterior to the circle $x^2+y^2-1=0$.

Similarly, in the formula

$$z_1=\frac{z-\alpha}{1-\bar{\alpha}z}\cdot\frac{z-\beta}{1-\bar{\beta}z},$$

the case considered ($\alpha=a=0$, $\beta=b=\sqrt{2}$) has been that of an interior point α, and an exterior point β. I omit the case of two exterior points, since this can be by the transformation $\dfrac{1}{z_1}$ for z_1 reduced to that of two interior points, but it is proper to consider this last case.

29. Considering then the case

$$z_1=\frac{z-\alpha}{1-\bar{\alpha}z}\cdot\frac{z-\beta}{1-\bar{\beta}z},$$

where α, β are interior points, the most simple case is when $\alpha=a$, $\beta=-a$, positive and <1. Here

$$z_1=\frac{z^2-a^2}{1-a^2z^2},$$

and we have

$$x_1+iy_1=\frac{(x^2-y^2)(x^2+y^2)-2a^2(x^2-y^2)+a^4}{1-2a^2(x^2-y^2)+a^4(x^2+y^2)^2},$$

or say

$$x_1^2+y_1^2-1=\frac{(1-a^4)\{(x^2+y^2)^2-1\}}{1-2a^2(x^2-y^2)+a^4(x^2+y^2)^2},$$

which last form shows that to the circumference $x^2+y^2-1=0$, there corresponds the circumference $x_1^2+y_1^2-1=0$ (and besides this, only the imaginary curve $x^2+y^2+1=0$). Writing $x_1^2+y_1^2=c^2$, we have the contour

$$(c^2a^4-1)(x^2+y^2)^2-2a^2(c^2-1)(x^2-y^2)+c^2-a^4=0,$$

a bicircular quartic. For $c=1$, we have as already mentioned the circle $x^2+y^2-1=0$; for c less than 1, a curve lying within this circle, diminishing with c, and after a time acquiring on each side of the axis of x an indentation or assuming an hour-glass form; for the value $c^2=a^4$, the equation becomes

$$(a^4+1)(x^2+y^2)^2-2a^2(x^2-y^2)=0,$$

and the curve is a figure of eight, the two loops enclosing the points $y=0$, $x=a$ and $x=-a$ respectively; and as c^2 diminishes to zero, the curve consists of two detached ovals lying within the two loops of the figure of eight, and ultimately reducing themselves to the two points respectively. There is no difficulty in finding the equation of the curves corresponding to a radius $x_1-\theta y_1=0$, but the configuration of these curves is at once seen from that of the former system. We may in the present case say that the circle is squarewise contracted to the figure of eight; and then further that each loop of the figure of eight is squarewise contracted to a point; but we do not have a squarewise contraction of the circle to a point.

30. A closed curve or contour may be squarewise contracted not into a point but into a finite line: we see this in the case of a system of confocal ellipses, which gives the contraction of an ellipse into the thin ellipse which is the finite line joining the two foci. There is a like contraction of the circle; this is, in fact, given by the formula due to Schwarz, "Ueber einige

Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120 (see p. 115) for the orthomorphosis of the ellipse into a circle: this is

$$x_1 + iy_1 = \operatorname{sn}\frac{2K}{\pi}(X + iY),$$

where, if $a^2 - b^2 = 1$ and $a = \cos\dfrac{i\pi K'}{4K}$, or, what is the same thing, $ib = \sin\dfrac{i\pi K'}{4K}$, then the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{a^2-1} = 1$ is transformed into the circle $x_1^2 + y_1^2 - \dfrac{1}{k} = 0$. The contours and trajectories for the ellipse are the confocal ellipses and hyperbolas respectively;

and for the circle, they are two sets of bicircular quartics, such that the portions within the circle have a configuration resembling that of the confocal ellipses and hyperbolas within the ellipse. To investigate the formulae, it is convenient to introduce the function

$$X + iY = \sin^{-1}(x + iy);$$

we then have

$$x + iy = \sin(X + iY),$$

or say

$$x = \sin X \cos iY, \qquad y = \cos X \sin iY,$$

so that to any given value of Y there corresponds the ellipse

$$\frac{x^2}{\cos^2 iY} + \frac{y^2}{-\sin^2 iY} = 1,$$

and then

$$x_1 + iy_1 = \operatorname{sn}\frac{2K}{\pi}(X + iY),$$

or if

$$s = \operatorname{sn}\frac{2KX}{\pi}, \qquad s_1 = \operatorname{sn}\frac{2KiY}{\pi} = \operatorname{sn}\frac{2KX}{\pi}\sqrt{(1-s^2)},$$

$$c = \operatorname{cn}\frac{2KX}{\pi} = \sqrt{(1-s^2)}, \qquad c_1 = \operatorname{cn}\frac{2KiY}{\pi} = \sqrt{(1+s_1^2)},$$

$$d = \operatorname{dn}\frac{2KX}{\pi} = \sqrt{(1-k^2s^2)}, \qquad d_1 = \operatorname{dn}\frac{2KiY}{\pi} = \sqrt{(1+k^2s_1^2)},$$

then

$$x_1 = \frac{sc_1d_1}{1+k^2s^2s_1^2}, \qquad y_1 = \frac{s_1cd}{1+k^2s^2s_1^2},$$

giving

$$x_1^2 + y_1^2 = \frac{s^2+s_1^2}{1+k^2s^2s_1^2},$$

and therefore $x_1^2 + y_1^2 - \dfrac{1}{k} = 0$ if $1 - ks_1^2 = 0$, that is, if $s_1 = \dfrac{1}{\sqrt{k}}$, or since $s_1 = \operatorname{sn}\dfrac{2KiY}{\pi}$, $\dfrac{2KiY}{\pi} = \frac{1}{2}iK'$, or $Y = \dfrac{\pi K'}{4K}$; hence defining a as above, it appears that the elliptic periphery $\dfrac{x^2}{a^2} + \dfrac{y^2}{a^2-1} = 1$ corresponds to the circumference $x_1^2 + y_1^2 - \dfrac{1}{k} = 0$. By the introduction of $X + iY$ as above, the circle and the ellipse are each compared with a rectangle; the reduction of the circle to the rectangle, as given by the foregoing equation $x_1 + iy_1 = \operatorname{sn}\dfrac{2K}{\pi}(X + iY)$, or what is substantially the same thing by an equation $x_1 + iy_1 = \operatorname{sn}(X + iY)$, is more fully discussed in my paper, Cayley: "On the Binodal Quartic and the graphical representation of the

Elliptic Functions," *Camb. Phil. Trans.*, t. XIV. (1889), pp. 484—494, [891], and in a paper "On some problems of orthomorphosis," *Crelle*, t. CVII. (1891), pp. 262—277, [921].

31. The whole theory, and in particular Riemann's theorem before referred to,

are intimately connected with Cauchy's theorem, "If a function $f(z)$ is holomorphic over a simply connected plane area, and if t denote any point within the area, then

$$f(t) = \frac{1}{2\pi i} \int \frac{f(z)}{z - t}\, dz,$$

where z denotes $x + iy$, and the integral is taken in the positive sense along the boundary of the area." See Briot and Bouquet, *Théorie des fonctions elliptiques* (Paris, 1875), p. 136.

Here in order to obtain by means of the theorem the value of the function $f(z)$ for a given point t $(= a + ib)$ within the area, we require to know the values of $f(z)$ for the several points of the boundary: viz. if z refers to a point P on the boundary, and if we represent the value $f(z)$ by a point P_1 in a second figure, then these points P_1 form a closed curve or boundary in this second figure, and we require to know not only the form of this boundary, but also the several points P_1 thereof which correspond to the several points P of the first-mentioned boundary, or say we require to know the correspondence of the two boundaries: this being known, we have by the theorem the value of $f(t)$, that is, the point $a_1 + ib_1$ within the second area, which corresponds to the point $t = a + ib$ within the first area. The $(1, 1)$ correspondence of the two areas is of course implied in the assertion that $f(t)$ has a determinate value, determined by means of the given values of $f(z)$ along the boundary.

ON SOME PROBLEMS OF ORTHOMORPHOSIS

[921]

From Crelle's Journal der Mathem., t. cvii. (1891), pp. 262—277.

In the interesting Memoir, Schwarz, "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120, the author considers the orthomorphic transformation (or, as I call it, the orthomorphosis) of a square into the infinite half-plane, or into a circle, and of a rectangle into the infinite half-plane. It is of course easy to deduce the orthomorphosis of the rectangle into a circle; and then, by giving a proper value to the modulus of the elliptic function involved in the formula, we obtain the orthomorphosis of the square into a circle, this solution (although equivalent thereto) being under a different form from that previously given for the square. But as appears from my paper "On the Binodal Quartic and the Graphical Representation of the Elliptic Functions," *Camb. Phil. Trans.*, vol. XIV. (1889), pp. 484—494, [891], there is for the rectangle (and consequently also for the square) an orthomorphosis wherein the boundary of the rectangle or square corresponds, not to the circumference, but to the circumference together with two twice-repeated portions of a diameter. I propose to consider in I. these several transformations: II. relates to the orthomorphosis of a circle into a circle.

I.

1. We are concerned with the elliptic function sn for which $K' = 2K$, and also with the elliptic function snl of the lemniscate form. The modulus in the former case is

$$k^2 = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2 = 3-2\sqrt{2}, \quad \text{or say } \sqrt{k} = \sqrt{2}-1, \quad \frac{1}{\sqrt{k}} = \sqrt{2}+1.$$

For the lemniscate function snl, the modulus is $= i$, and if (with Gauss) we write $\frac{1}{2}\varpi$ for the value of the complete function K or F_1, then we have $\frac{1}{2}\varpi = K(\sqrt{2}-1)$,
$= K\sqrt{k}$, where k has the foregoing special value: I notice the numerical values $\frac{1}{2}\varpi = 1.311028$, $k = 3-2\sqrt{2} = \sin 9°52'$, $\frac{1}{\sqrt{k}} = 1.553774$. The relation between the lemniscate function snl, and the sn with the foregoing value of k, is easily shown to be

$$\sqrt{k}\,\operatorname{sn} U = \frac{(i+1)\operatorname{snl} u}{(i-1)\operatorname{snl} u + \sqrt{2}}, \quad \text{where} \quad U = \tfrac{1}{2}(K' - iK) + i\sqrt{2}\,\frac{u}{\sqrt{k}}, \quad u = \frac{1+i}{\sqrt{2}}\,U,$$

and it may be added that we have

$$\operatorname{cn} U = \frac{\sqrt{2}}{(i-1)\operatorname{snl} u + \sqrt{2}} \sqrt{\frac{1+k}{k}}\,\sqrt{(1+\operatorname{snl} u)(1-i\operatorname{snl} u)},$$

$$\operatorname{dn} U = \frac{\sqrt{2}}{(i-1)\operatorname{snl} u + \sqrt{2}} \sqrt{1+k}\,\sqrt{(1-\operatorname{snl} u)(1+i\operatorname{snl} u)}.$$

2. The Schwarzian orthomorphosis of the rectangle into the infinite half-plane is given (Memoir, p. 113) by the formula $X_1 + iY_1 = \operatorname{sn}(X+iY)$, where the modulus is real, positive, and less than unity. Here, see the figures 1 (XY) and 2 (X_1Y_1),

Fig. 1 (XY).

Fig. 2 (X_1Y_1).

the rectangle $ABCD$, the sides of which are $AB = 2K$ and $BC = K'$, is transformed into the upper infinite half-plane ($Y_1 = +$), the four corners of the rectangle corresponding to the points A, B, C, D on the axis of X_1, where $OB(= OA) = 1$, $OC(= OD) = \dfrac{1}{k}$.

3. We can, by a properly determined quasi-inversion (as will be explained), transform the X_1Y_1-figure into a new figure see figure 3 (X_2Y_2), the infinite X_1-axis being transformed into the circumference of a circle (the radius of which may be taken to be $= 1$) and the infinite half-plane into the area within the circle. The four points A, B, C, D are thus transformed into points on the circle, which if the quasi-inversion be a symmetrical one, will be situate, A and B symmetrically, and

also C and D symmetrically, in regard to the axis OY_2: and in the case of the before-mentioned modulus $k = 3 - 2\sqrt{2}$, for which the rectangle $ABCD$ becomes a

Fig. 3 (X_2Y_2).*

square, the quasi-inversion may be so determined that the points A, B, C, D shall be situate midway (that is, at inclinations $\pm 45°$, $\pm 135°$) in the four quadrants of the circle.

4. But if in figure 1 we take $EL = \frac{1}{2}K'$ and draw FL parallel to OX, then as shown in my memoir above referred to, the foregoing transformation $X_1 + iY_1 = \mathrm{sn}(X + iY)$ changes the rectangle $OBLF$, the sides of which are $OB = K$ and $BL = \frac{1}{2}K'$, into the quadrant $OBLF$ of figure 2, $OB = 1$ (as already mentioned) and radius $OL = \dfrac{1}{\sqrt{k}}$. Hence completing the rectangle $RSLM$, the sides of which are $RS = 2K$ and $SL = K'$, (viz. this rectangle differs only in position from the first-mentioned rectangle $ABCD$), we have an orthomorphosis of the rectangle $RSLM$ into the circle of figure 2: but so nevertheless that to the boundary of the rectangle there corresponds not the circumference of the circle but the boundary $RGSBLFMAR$ composed of the circumference and the portions SB, BL and MA, AR (that is, BL, AM each twice) of a diameter of the circle. See post No. 12.

5. The Schwarzian orthomorphosis of the square into a circle is given (Memoir, pp. 111—113) by the formula $x_1 + iy_1 = \mathrm{snl}(x + iy)$; viz. here—see figures 4 (xy) and 5 (x_1y_1)—the square $ABCD$, the half-diagonals whereof are each $= \frac{1}{2}\varpi$, corresponds to the circle radius $= 1$ of figure 5, the four points A, B, C, D of the circle being the quadrantal points as shown in the figure. Figure 5 is, in fact, figure 3 turned through an angle of 45°; and it thus appears that Schwarz's lemniscate solution for the square into a circle is the quasi-inversion of his solution for the rectangle into the infinite half-plane, when by putting $k = 3 - 2\sqrt{2}$ the rectangle is made to be a square. See post Nos. 13 and 14.

6. The general formula of quasi-inversion whereby the infinite X_1-axis is changed

Fig. 4 (xy). *Fig. 5* (x_1y_1).*

into a circumference[2], radius unity, is

$$X_2 + iY_2 = \frac{1 + Mi(X_1 + iY_1)}{M(X_1 + iY_1) + i},$$

where M is real. In fact, writing this equation in the form

$$X_2 + iY_2 = \frac{1 + Mi(X_1 + iY_1)}{i\{1 + Mi(X_1 - iY_1)\}^{-1} \cdot M(X_1 + iY_1) + i},$$

we have

$$X_2 - iY_2 = \frac{1 - Mi(X_1 - iY_1)}{-i\{1 + Mi(X_1 - iY_1)\}},$$

[1]* The scale of this figure is double that of figures 1, 2, 4, 6, 7 in this paper.

[2]I use here and elsewhere the term circumference rather than circle, to mark more clearly the distinction between the curve and the included area.

and thence $X_2^2+Y_2^2-1=0$, if only $Y_1=0$; that is, the infinite X_1-axis is transformed into the circumference $X_2^2+Y_2^2-1=0$.

Writing $Y_1=0$, we have

$$X_2+iY_2=\frac{1+MiX_1}{MX_1+i}=\frac{2MX_1+i(M^2X_1^2-1)}{M^2X_1^2+1},$$

so that to a pair of points $(\pm X_1, 0)$ there corresponds a pair of points $(\mp X_2, Y_2)$ situate symmetrically in regard to the axis OY_2.

The equation gives

$$M(X_1+iY_1)=\frac{1-i(X_2+iY_2)}{-i\{1+i(X_2-iY_2)\}^{-1}},$$

Hence, writing this in the form

$$M(X_1+iY_1)=\frac{1-i(X_2+iY_2)}{i\{1+i(X_2+iY_2)\}},$$

we have

$$M(X_1-iY_1)=\frac{1+i(X_2-iY_2)}{i\{1-i(X_2-iY_2)\}},$$

and consequently $M^2(X_1^2+Y_1^2)-\dfrac{1}{M^2}=0$, if only $Y_2=0$; that is, the circumference $X_1^2+Y_1^2-\dfrac{1}{M^2}=0$ is transformed into the infinite X_2-axis.

Although for the purposes of the present memoir we require the coefficient M, yet there is no real loss of generality in assuming $M=1$, and the transformation is best studied under the form

$$X_2+iY_2=\frac{1+i(X_1+iY_1)}{(X_1+iY_1)+i};$$

see post No. 11.

7. As already mentioned, the coordinates (X_1, Y_1) and (X_2, Y_2) are connected by an equation of the foregoing form, and in the case of the square ($k=3-2\sqrt{2}$ as before), the corresponding values for the points A, B, C, D should be

$$\begin{array}{llll}
A, & X_1+iY_1=-1, & X_2+iY_2=\dfrac{-1-i}{\sqrt{2}}, \\
B, & X_1+iY_1=1, & X_2+iY_2=\dfrac{1-i}{\sqrt{2}}, \\
C, & X_1+iY_1=\dfrac{1}{k}, & X_2+iY_2=\dfrac{1+i}{\sqrt{2}}, \\
D, & X_1+iY_1=-\dfrac{1}{k}, & X_2+iY_2=\dfrac{-1+i}{\sqrt{2}}.
\end{array}$$

The proper value of M is $M=\sqrt{2}-1=\sqrt{k}$; the required formula thus is

$$\sqrt{k}(X_1+iY_1)=\frac{1+i\sqrt{k}(X_2+iY_2)}{\sqrt{k}(X_2+iY_2)+i},$$

or say

$$\sqrt{k}(X_1+iY_1)=\frac{1-i(X_2+iY_2)}{X_2+iY_2-i}.$$

2* The scale of this figure is double that of figures 1, 2, 4, 6, 7 in this paper.

Thus for the point B, we should have

$$\sqrt{k} = \frac{1 - i \cdot \dfrac{1-i}{\sqrt{2}}}{\dfrac{1-i}{\sqrt{2}} - i}, \quad \text{that is,} \quad \sqrt{k} = \frac{\sqrt{2} - i - 1}{1 - i\sqrt{2} - i} = \frac{\sqrt{2} - 1 - i}{1 - i - i\sqrt{2}},$$

which is right. And similarly for the point C, we should have

$$\frac{1}{\sqrt{k}} = \frac{1 - i \cdot \dfrac{1+i}{\sqrt{2}}}{\dfrac{1+i}{\sqrt{2}} - i}, \quad \text{that is,} \quad \frac{1}{\sqrt{k}} = \frac{\sqrt{2} - i + 1}{1 + i - i\sqrt{2}} = \sqrt{2} + 1,$$

which is right. And similarly for the points A and D.

8. We connect the square $ABCD$ of figure 1 with that of figure 4 by the equation

$$X + iY - \tfrac{1}{2}iK' = \frac{1+i}{2\sqrt{k}}(x + iy);$$

in fact, recollecting that $\frac{1}{2}K' = K = \dfrac{\varpi}{4\sqrt{k}}$, we have for the four points respectively

$$\begin{array}{lll} A, & X + iY = -K, & x + iy = -\tfrac{1}{2}\varpi, \\ B, & X + iY = K, & x + iy = -\tfrac{1}{2}\varpi i, \\ C, & X + iY = K + iK', & x + iy = \tfrac{1}{2}\varpi, \\ D, & X + iY = -K + iK', & x + iy = \tfrac{1}{2}\varpi i, \end{array}$$

values which satisfy the relation in question.

9. Hence writing

$$X + iY = U, \quad x + iy = u,$$

we have

$$U - \tfrac{1}{2}iK' = \frac{1+i}{2\sqrt{k}}\, u,$$

which is the relation in No. 1 between the arguments U, u of the elliptic functions sn, snl. We have $X_1 + iY_1 = \operatorname{sn} U$, $x_1 + iy_1 = \operatorname{snl} u$; and consequently

$$\sqrt{k}(X_1 + iY_1) = \frac{(i+1)(x_1 + iy_1) + i\sqrt{2}}{(i-1)(x_1 + iy_1) + \sqrt{2}}.$$

Hence, substituting for $\sqrt{k}(X_1 + iY_1)$ its value in terms of $X_2 + iY_2$, we find

$$\frac{1 - i(X_2 + iY_2)}{X_2 + iY_2 - i} = \frac{(i+1)(x_1 + iy_1) + i\sqrt{2}}{(i-1)(x_1 + iy_1) + \sqrt{2}},$$

an equation which (multiplying on the left-hand side the numerator and the denominator each by $\dfrac{-i}{\sqrt{2}}$) may be written

$$\frac{1 - i + \dfrac{1+i}{\sqrt{2}}(x_1 + iy_1)}{\dfrac{1+i}{\sqrt{2}}(x_1 + iy_1) - \dfrac{1-i}{\sqrt{2}}} = \frac{1 - i(X_2 + iY_2)}{X_2 + iY_2 - i},$$

that is, we have

$$X_2 + iY_2 = \frac{1+i}{\sqrt{2}}(x_1 + iy_1),$$

an equation which shows that the figure 5 is in fact the figure 3 turned through an angle of 45°. We have thus proved the conclusion stated in No. 5 as to the connexion between the lemniscate solution for the square and the solution for the rectangle.

10. It is convenient to collect here the several equations relating to the orthomorphosis of the square. We have

$$X_1 + iY_1 = \operatorname{sn}(X + iY), \quad k = 3 - 2\sqrt{2};$$

$$x_1 + iy_1 = \operatorname{snl}(x + iy);$$

$$\sqrt{k}(X_1 + iY_1) = \frac{1 - i(X_2 + iY_2)}{X_2 + iY_2 - i};$$

$$X + iY - \tfrac{1}{2}iK' = \frac{1+i}{2\sqrt{k}}(x + iy);$$

$$\sqrt{k}(X_1 + iY_1) = \frac{(i+1)(x_1 + iy_1) + i\sqrt{2}}{(i-1)(x_1 + iy_1) + \sqrt{2}};$$

$$X_2 + iY_2 = \frac{1+i}{\sqrt{2}}(x_1 + iy_1),$$

which are the equations connecting together the coordinates of the five figures.

11. I examine more in detail the above-mentioned transformation

$$X_2 + iY_2 = \frac{1 + i(X_1 + iY_1)}{(X_1 + iY_1) + i};$$

see the foregoing figures 1 (XY) and 2 (X_1Y_1), in which we now regard the two circles as having each of them the radius unity; changing the sign of i, the equation gives

$$X_2 - iY_2 = \frac{1 - i(X_1 - iY_1)}{(X_1 - iY_1) - i},$$

and we hence find

$$X_2^2 + Y_2^2 + 1 = \frac{2(X_1^2 + Y_1^2 + 1)}{X_1^2 + Y_1^2 + 2Y_1 + 1};$$

consequently if $X_1^2 + Y_1^2 + 1 = 0$, then also $X_2^2 + Y_2^2 + 1 = 0$, or the transformation changes the first of these imaginary circles into the second of them: or say it changes the concentric orthotomic of the circle $X_1^2 + Y_1^2 - 1 = 0$ into the concentric orthotomic of the circle $X_2^2 + Y_2^2 - 1 = 0$.

We have moreover

$$X_2^2 + Y_2^2 - 1 = \frac{X_1^2 + Y_1^2 - 1 - 2Y_1}{X_1^2 + Y_1^2 + 2Y_1 + 1}, \qquad Y_2 = \frac{Y_1 - 1 \cdot \frac{1}{2}(X_1^2 + Y_1^2 - 1)}{X_1^2 + Y_1^2 + 2Y_1 + 1},$$

values which give the foregoing expression for $X_2^2 + Y_2^2$. But we further obtain

$$X_2^2 + Y_2^2 - 1 - 2\mu Y_2 = \frac{X_1^2 + Y_1^2 - 1 + \dfrac{2}{\mu}Y_1}{X_1^2 + Y_1^2 + 2Y_1 + 1} \cdot (1 - \mu),$$

and it thus appears that the circumferences

$$X_1^2 + Y_1^2 - 1 + 2\mu Y_1 = 0, \qquad X_2^2 + Y_2^2 - 1 - \frac{2}{\mu}Y_2 = 0,$$

correspond to each other. These are circles passing through the pairs of points $(Y_1 = 0, X_1 = \pm 1)$, $(Y_2 = 0, X_2 = \pm 1)$ respectively; or imagining them in the same figure, say they are circles which belong each to the series of circles $x^2 + y^2 - 1 + 2\beta y = 0$, and which moreover cut at right angles at the points $y = 0$, $x = \pm 1$. But attending more carefully to the nature of the correspondence, it is to be observed that taking Y_1 positive, and giving to X_1 any positive value from 0 to ∞, we have in figure 2 an arc LJM lying wholly within the upper semicircle LFM; and that, corresponding hereto in figure 3, we have an arc LJM lying wholly within the lower semicircle LOM; and that as in figure 2, the arc LJM lies nearer to the semicircumference LFM or to the diameter LOM, so in figure 3 the arc LJM lies nearer to the diameter LFM or to the semicircumference LOM. Thus the upper semicircle LFM of figure 2 corresponds to the lower semicircle LOM of figure 3; but so that the semicircumference LFM of the first figure corresponds to the diameter LFM of the second figure; and the diameter LOM of the first figure to the semicircumference LOM of the second figure. And further supposing that Y_1 is still positive, but that μ has any negative value from $-\infty$ to 0, we have in figure 2 an arc in the upper half-plane lying wholly outside the semicircle; and corresponding thereto in figure 3, an arc lying wholly inside the upper semicircle LHM; that is, in figure 2 the infinite space in the upper half-plane outside the semicircle corresponds in figure 3 to the space within the upper half-circle LHM; the infinity of figure 2 corresponding to the semicircumference LHM of figure 3, and the semicircumference LFM of figure 2 to the diameter LFM of figure 3. And thus in figure 2, the upper half-plane inside and outside the semicircle corresponds in figure 3 to the lower and upper half-circles, that is, to the whole circular area $OLHM$ of figure 3.

It is to be observed, moreover, that we have

$$X_2^2 + Y_2^2 + 1 + 2\lambda X_2 = \frac{2(X_1^2 + Y_1^2 + 1 + 2\lambda X_1)}{X_1^2 + Y_1^2 + 2Y_1 + 1},$$

that is, the circles $X_1^2 + Y_1^2 + 1 + 2\lambda X_1 = 0$ and $X_2^2 + Y_2^2 + 1 + 2\lambda X_2 = 0$ correspond to each other. Imagining the circles as belonging to the same figure, these are one and the same circle of the series $x^2 + y^2 + 1 - 2\alpha x = 0$ each passing through the pair of points $(x = 0, y = \pm i)$ which are the antipoints of the pair $(y = 0, x = \pm 1)$; these circles thus cut at right angles those of the series $x^2 + y^2 - 1 + 2\beta y = 0$. We have, by means of the two series of circles, an easy construction for the correspondence between the figures 2 and 3.

12. In explanation of No. 4, observe that, starting from the equation

$$X_1 + iY_1 = \operatorname{sn}(X + iY),$$

and writing $\operatorname{sn} X = P$, $\operatorname{sn} iY = iQ$, we have

$$X_1 + iY_1 = \frac{P\sqrt{1+Q^2}\sqrt{1+k^2Q^2} + iQ\sqrt{1-P^2}\sqrt{1-k^2P^2}}{1 + k^2P^2Q^2},$$

that is,

$$X_1 = \frac{P\sqrt{1+Q^2}\sqrt{1+k^2Q^2}}{1 + k^2P^2Q^2}, \qquad Y_1 = \frac{Q\sqrt{1-P^2}\sqrt{1-k^2P^2}}{1 + k^2P^2Q^2},$$

and thence

$$X_1^2 + Y_1^2 = \frac{P^2 + Q^2}{1 + k^2P^2Q^2} = r^2 \quad (\text{if } X_1^2 + Y_1^2 \text{ be put } = r^2);$$

hence

$$P^2(1 - k^2Q^2r^2) = r^2 - Q^2, \qquad Q^2(1 - k^2P^2r^2) = r^2 - P^2.$$

Now considering in figure 1 any line in the rectangle parallel to the axis of X, that is, taking Y constant and therefore also Q constant, and proceeding to eliminate P, we have

$$1 - P^2 = \frac{1 + Q^2 - (1 + k^2Q^2)r^2}{1 - k^2Q^2r^2}, \qquad 1 - k^2P^2 = \frac{1 + k^2Q^2 - k^2(1 + Q^2)r^2}{1 - k^2Q^2r^2},$$

and consequently

$$Y_1^2 = \frac{Q^2\{1+Q^2-(1+k^2Q^2)r^2\}\{1+k^2Q^2-k^2(1+Q^2)r^2\}}{(1-k^2Q^2r^2)^2},$$

giving X_1, Y_1 each of them in terms of Q^2 and r^2, $= X_1^2+Y_1^2$. The former of these equations, replacing therein r^2 by its value, gives easily

$$(X_1^2+Y_1^2)^2 - 2AX_1^2 - 2BY_1^2 + \frac{1}{k^2} = 0,$$

where

$$2A = \frac{1+Q^2}{1+k^2Q^2} + \frac{1}{k^2}\cdot\frac{1+k^2Q^2}{1+Q^2}, \qquad 2B = Q^2 + \frac{1}{k^2Q^2};$$

viz. we have thus the equation of the curve, a bicircular quartic which in figure 2 corresponds to the line parallel to the axis of X_1 in figure 1.

In particular, for the line LM of figure 1 we have $Y = \frac{1}{2}K'$ and thence $iQ = \mathrm{sn}\,\frac{1}{2}iK' = \dfrac{i}{\sqrt{k}}$, that is, $Q = \dfrac{1}{\sqrt{k}}$, and thence $A = B = \dfrac{1}{k}$; the equation of the bicircular quartic is

$$(X_1^2+Y_1^2)^2 - \frac{2}{k}(X_1^2+Y_1^2) + \frac{1}{k^2} = 0,$$

viz. this is the circle $X_1^2+Y_1^2-\dfrac{1}{k}=0$ twice repeated, and we have thus this circle, or rather the half-circumference LFM of figure 2, corresponding to the line LM of figure 1. More simply,

$$X_1 + iY_1 = \frac{1}{\sqrt{k}}\,\mathrm{sn}\,X + i\,\frac{i}{\sqrt{k}}\cdot\mathrm{cn}\,X\ \mathrm{dn}\,X,$$

and thence

$$X_1^2+Y_1^2 = \frac{\mathrm{sn}^2 X + (\mathrm{cn}\,X\ \mathrm{dn}\,X)^2/k^2}{1+k\ \mathrm{sn}^2 X\cdot(1/k)} = \frac{1}{k},$$

that is, $X_1^2+Y_1^2-\dfrac{1}{k}=0$ as before. It is easy to see that the points A, O, B of figure 1 correspond to the points A, O, B of figure 2, and hence that the area of the rectangle $AOBLFMA$ of figure 1 corresponds to that of the semicircle $AOBLFMA$ of figure 2.

Returning to the equation $X_1+iY_1 = \mathrm{sn}(X+iY)$, if we write herein successively

$$Y_1 = \tfrac{1}{2}K'+\beta, \quad \mathrm{sn}\,iY_1 = iQ_1 = \mathrm{sn}\,i(\tfrac{1}{2}K'+\beta),$$

and

$$Y_2 = \tfrac{1}{2}K'-\beta, \quad \mathrm{sn}\,iY_2 = iQ_2 = \mathrm{sn}\,i(\tfrac{1}{2}K'-\beta),$$

then we have

$$Q_1Q_2 = \frac{1}{k};$$

hence for q writing Q_1 or Q_2, we have in each case the same values of A and B, that is, we have the same bicircular quartic for two lines parallel to and equidistant from the line LM, but to one of these (viz. the line between LM and BA) there corresponds the half-perimeter lying within the semicircumference LFM, and to the other of them (viz. the line between LM and CD) there corresponds the half-perimeter lying without the semicircumference LFM.

It may be shown in like manner that, to any line in figure 1 parallel to the axis OY, there corresponds in figure 2 a bicircular quartic of the like form

$$(X_1^2+Y_1^2)^2 - 2AX_1^2 - 2BY_1^2 + \frac{1}{k^2} = 0.$$

13. Similarly in explanation of No. 5, observe that, starting from the equation $x_1 + iy_1 = \operatorname{snl}(x+iy)$ and writing $\operatorname{snl} x = p$, $\operatorname{snl} iy = i \operatorname{snl} y = iq$, we have

$$x_1 + iy_1 = \frac{p\sqrt{1-q^4} + iq\sqrt{1-p^4}}{1-p^2q^2},$$

that is,

$$x_1 = \frac{p\sqrt{1-q^4}}{1-p^2q^2}, \qquad y_1 = \frac{q\sqrt{1-p^4}}{1-p^2q^2},$$

and thence

$$x_1^2 + y_1^2 = \frac{p^2+q^2}{1-p^2q^2};$$

writing $x + y = \frac{1}{2}\varpi$, we have $\operatorname{snl} y = \operatorname{cnl} x$, that is, $q = \dfrac{\sqrt{1-p^2}}{\sqrt{1+p^2}}$, or $q^2 = \dfrac{1-p^2}{1+p^2}$, that is, $\dfrac{p^2}{1+p^2} + \dfrac{q^2}{1-p^2} = 1$; and thus to the line $x+y = \frac{1}{2}\varpi$, there corresponds the circle $x_1^2+y_1^2-1=0$. More precisely, to the line CD of figure 4, there corresponds the quarter-circumference CD

of figure 5: and similarly to DA, AB and BC of figure 4 the remaining quarter-circumferences DA, AB, BC of figure 5; that is, to the whole boundary of the square in figure 4 there corresponds the whole circumference of the circle in figure 5.

14. To any line in figure 4, parallel to the axis Ox or to the axis Oy, there corresponds in figure 5 a bicircular quartic of the form $x_1^4 + y_1^4 - 2Ax_1^2 - 2By_1^2 - 1 = 0$. The investigation is substantially the same as that contained in No. 12, and need not be here given. But it is remarkable that also, to any line of figure 4 parallel to a side of the square (that is, to any line $x \pm y = c$), there corresponds in figure 5 a bicircular quartic of the like form (for the sides of the square, or lines $x \pm y = \pm\frac{1}{2}\varpi$ of figure 4, this bicircular quartic becomes the twice repeated circle $x_1^2 + y_1^2 - 1 = 0$ of figure 5, which is the result just obtained in No. 13). I investigate this result as follows. Writing, as in No. 13,

$$\operatorname{snl} x = p, \quad \operatorname{snl} iy = i \operatorname{snl} y = iq,$$

we have, as above,

$$x_1 = \frac{p\sqrt{1-q^4}}{1-p^2q^2}, \qquad y_1 = \frac{q\sqrt{1-p^4}}{1-p^2q^2}, \quad \text{and thence} \quad x_1^2 + y_1^2 = \frac{p^2+q^2}{1-p^2q^2}.$$

Now assuming between x and y the relation $x + y = C$, and writing $\operatorname{snl} C = c$, this gives

$$c = \frac{p\sqrt{1-q^4} + q\sqrt{1-p^4}}{1-p^2q^2};$$

to obtain the required curve, we must between these equations (three independent equations) eliminate p and q. We have

$$x_1 + y_1 = \frac{p\sqrt{1-q^4} + q\sqrt{1-p^4}}{1-p^2q^2},$$

and consequently

$$(1+p^2q^2)c = (1-p^2q^2)(x_1+y_1),$$

or, writing for convenience $\Omega = 1 - p^2q^2$, this equation gives

$$\Omega = \frac{2c}{x_1+y_1+c}.$$

Hence $\Omega x_1 = p\sqrt{1-q^4}$, $\Omega y_1 = q\sqrt{1-p^4}$; these equations may be written

$$\Omega^2 x_1^2 = p^2(1-q^2)(1+q^2), \qquad \Omega^2 y_1^2 = q^2(1-p^2)(1+p^2),$$

and from these equations obtaining the expressions for p^2 and q^2, and thence the expression for p^2q^2, $= 1-\Omega$, we find, after some easy reductions,

$$(x_1^2+y_1^2)^2 - \frac{4c^2x_1^2y_1^2}{(x_1+y_1+c)^2} = (1-c^2)\,\Omega.$$

But we have

$$\Omega = \frac{2c}{x_1+y_1+c}, \quad \text{or} \quad \Omega = \frac{2c}{P+2Q\,c} \quad (P = x_1^2+y_1^2,\ Q = x_1y_1);$$

or substituting this value, the equation becomes

$$(x_1^2+y_1^2)^2 - \frac{4c^2x_1^2y_1^2}{(x_1+y_1+c)^2} = \frac{2c(1-c^2)}{x_1+y_1+c},$$

that is,

$$\left\{(x_1^2+y_1^2)^2(x_1+y_1+c)^2 - 4c^2x_1^2y_1^2\right\}(x_1+y_1+c) - 4c(x_1^2+y_1^2)(1-c^2)(x_1+y_1+c)^2 = 0.$$

Writing for a moment $x_1^2+y_1^2 = P$, $x_1y_1 = Q$, this is

$$\left\{P - \frac{1}{c^2}(P+2Q)\right\}\{P+2Q-c^2\} + 4c^2Q^2 = 0,$$

an equation which contains the factor $P+2Q$; throwing this out, the equation becomes, after an easy reduction,

$$(P-1)^2 + (1-c^2)\left\{2p_1^2 - \frac{1}{c^2}(P+2Q)\right\} = 0,$$

that is,

$$(x_1^2+y_1^2-1)^2 + (1-c^2)\left\{(x_1-y_1)^2 - \frac{1}{c^2}(x_1+y_1)^2\right\} = 0,$$

the required equation. Transforming through an angle of 45° by writing

$$x_1 = \frac{x_2+y_2}{\sqrt{2}}, \qquad y_1 = \frac{x_2-y_2}{\sqrt{2}},$$

(where observe that the axis of x_2 is the line FH of figure 5), the equation becomes

$$(x_2^2+y_2^2-1)^2 + (1-c^2)\,2y_2^2 - \frac{2}{c^2}\,x_2^2 = 0,$$

or writing $c = \cos\gamma$ and therefore $1-c^2 = \sin^2\gamma$, this equation becomes

$$(x_2^2+y_2^2)^2 - \frac{2}{\cos^2\gamma}\,x_2^2 - 2\cos^2\gamma\cdot y_2^2 + 1 = 0,$$

a curve consisting of two indented ovals situate symmetrically in regard to the axis Fy_2 of figure 5. In fact, writing in the equation $y_2 = 0$, we have for x_2^2 two real positive values; but, writing $x_2 = 0$, we have for y_2^2 two imaginary values. For $\gamma = 0$, the equation becomes

$$(x_2^2+y_2^2)^2 - 2(x_2^2+y_2^2) + 1 = 0,$$

that is, we have the circle $x_2^2+y_2^2-1 = 0$ twice repeated. One of the ovals is shown in figure 5; the portion of it lying within the circle agrees with Schwarz's figure, p. 113, turning this round through an angle of 45°.

For the lines $x-y = C$ of figure 4, we have in figure 5 the same system of bicircular quartics turned round through an angle of 90°.

II.

15. I consider the general problem of the orthomorphosis of a circle into a circle: we can, for the transformation of the circumference of the circle $x^2 + y^2 - 1 = 0$ into that of the circle $x_1^2 + y_1^2 - 1 = 0$, find a formula involving an arbitrary function

or (what is the same thing) an indefinite number of arbitrary constants. In fact, writing for shortness $z = x + iy$, $z_1 = x_1 + iy_1$, and $\bar{z}$, $\bar{z}_1$ for the conjugate functions $x - iy$, $x_1 - iy_1$; also $\phi(z)$ for a function of z involving in general imaginary coefficients $a + ib$, &c., and $\bar{\phi}(z)$ for the like function with the conjugate coefficients $a - ib$, &c.; then if we assume

$$z_1 = \frac{\phi(z)}{z^m \bar{\phi}\left(\frac{1}{z}\right)},$$

where m is any positive or negative integer, this implies

$$\bar{z}_1 = \frac{\bar{\phi}(\bar{z})}{\bar{z}^m \phi\left(\frac{1}{\bar{z}}\right)};$$

consequently, if $x^2 + y^2 - 1 = 0$, that is, $z\bar{z} = 1$, or $\bar{z} = \frac{1}{z}$, we have

$$\bar{z}_1 = \frac{\bar{\phi}\left(\frac{1}{z}\right)}{\left(\frac{1}{z}\right)^m \phi(z)} = \frac{z^m\, \bar{\phi}\left(\frac{1}{z}\right)}{\phi(z)} = \frac{1}{z_1},$$

or $z_1\bar{z}_1 = 1$, that is, $x_1^2 + y_1^2 - 1 = 0$.

In a slightly different form, taking α, β, &c., to denote any imaginary quantities, and $\bar{\alpha}$, $\bar{\beta}$, ... the conjugate quantities; assuming

$$\phi(z) = (z - \alpha)(z - \beta)\ldots,$$

and taking m for the number of factors, we have

$$z_1 = \frac{(z-\alpha)(z-\beta)\ldots}{(1-\bar{\alpha}z)(1-\bar{\beta}z)\ldots},$$

and then (repeating the demonstration) we have

$$\bar{z}_1 = \frac{(\bar{z}-\bar{\alpha})(\bar{z}-\bar{\beta})\ldots}{(1-\alpha\bar{z})(1-\beta\bar{z})\ldots},$$

which, writing therein $\bar{z} = \frac{1}{z}$, becomes

$$\bar{z}_1 = \frac{\left(\frac{1}{z} - \bar{\alpha}\right)\left(\frac{1}{z} - \bar{\beta}\right)\ldots}{\left(1 - \frac{\alpha}{z}\right)\left(1 - \frac{\beta}{z}\right)\ldots} = \frac{1}{z_1},$$

and consequently, if $\bar{z} = \frac{1}{z}$, then also $\bar{z}_1 = \frac{1}{z_1}$ as before.

We may in the expression for z_1 introduce a factor $\frac{A}{\bar{A}}$, or, what is the same thing, a factor A which is such that $A\bar{A} = 1$. In particular, we thus have the solution

$$z_1 = \frac{A(z-\alpha)}{(1-\bar{\alpha}z)}.$$